Siluni Keerthiratne

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About Me

Machine Learning Engineer with hands-on experience building end-to-end AI solutions across computer vision and NLP domains, comfortable working with everything from traditional ML pipelines to agentic AI tools. I excel at translating business requirements into practical, scalable technical implementations, taking ownership from problem definition through deployment.

Key Skills and Qualifications

Programming & ML: LLM integration, agentic AI (LangGraph/LangChain), machine vision (YOLO family), data cleaning, feature engineering, and AI/ML workflow evaluation pipelines (LLM-as-judge, custom metrics).

Development: Backend: Python, FastAPI, and .NET Core; Frontend: React/Next.js.

Libraries & Tools: scikit-learn, TensorFlow, PyTorch, pandas, NumPy, Jupyter, Git, Docker, SQL, EF Core, Mediator pattern, CQRS, Clean Architecture.

Cloud & MLOps: AWS (Lambda, Bedrock, SQS, S3, Serverless Framework), Azure Machine Learning, PostgreSQL/pgvector.

Experience

Machine Learning Engineer, Ascentic (Pvt) Ltd. — June 2026 - Present

Associate Machine Learning Engineer, Ascentic (Pvt) Ltd. — July 2025 - June 2026

Software Engineer Intern, Ascentic (Pvt) Ltd. — July 2024 - June 2025

National Volunteer, United Nations World Food Programme — May 2022 - September 2022

Projects

Ascentic (Pvt) Ltd - Fintech Product Suite

Fintech: AI-Powered Invoice & Accounting Automation

- Owned end-to-end architecture and delivery of ML and agentic AI features for a large-scale fintech client, translating business requirements into production systems used by several Fortune 500 companies for e-invoicing and financial automation.
- Built an automated transaction-posting pipeline combining a LangGraph-based agentic workflow with per-tenant ML classifiers and a custom graph-based rule-mining engine, including an incremental/online learning loop, to classify and post financial transactions at scale.
- Built a serverless PDF-to-PEPPOL invoice extraction pipeline (AWS Lambda, Bedrock/Claude, Textract) converting unstructured PDFs and images into structured, standardized UBL e-invoice data, with OCR-based hallucination detection to validate LLM output before it reached production.
- Built a serverless multi-agent finance analyst platform (AWS Lambda, FastAPI, LangGraph, Bedrock/Claude) enabling natural-language financial analysis via a Text-to-SQL agent with self-correcting query refinement, stateful multi-turn sessions with Postgres-backed checkpointing, and SSE-streamed responses.
- Built a UNSPSC hyper-classification system using Claude-based enrichment, multilingual embeddings, and vector similarity search (pgvector) over a hierarchical taxonomy to automatically categorize financial line-item data at scale.
- Designed evaluation pipelines and an LLM-as-judge framework across the above products, tailoring customer-facing and traditional ML evals to validate model performance before and after deployment.
- **Tools Used**: Python, AWS Lambda, Bedrock, Textract, AppSync, SQS, S3, Aurora/RDS PostgreSQL, LangGraph/LangChain, scikit-learn, pgvector, FastAPI, Serverless Framework, Docker, GitHub Actions, Git.

Ascentic (Pvt) Ltd - Partnership Product

Medical Imaging and Computer Vision Model Development

- Trained and fine-tuned YOLO-based object detection models through iterative data cleaning, feature engineering, and hyperparameter optimization workflows.
- Enhanced dataset quality through image augmentation, labeling refinement, and class balancing to improve model generalization and performance.
- Leveraged Azure Machine Learning Studio for end-to-end model training, experiment management, and cloud-based pipeline orchestration.
- Integrated Comet ML for experiment tracking and comparative model evaluation, following MLOps best practices.
- **Tools Used**: Python, PyTorch, pandas, NumPy, YOLOv5-YOLOv12, Azure ML, Comet ML, Git.

United Nations World Food Programme - Projects

- Delivered Machine Learning and Python workshops for technical officers at the Department of Sensors and Statistics, Sri Lanka.
- Conducted data cleaning to ensure high-quality datasets for World Food Programme projects.
- Led survey data collection initiatives supporting World Food Programme operations.
- **Tools Used**: KuboToolbox

Education

Robert Gordon University, Aberdeen, Scotland September 2022 - April 2026

BSc Artificial Intelligence and Data Science (First Class Honours)

Musaeus College, Colombo 7, Sri Lanka January 2006 - March 2022

GCE Advanced Level Mathematics

Certifications

- Knowledge Graphs for RAG - DeepLearning.AI
- Neo4j Fundamentals - Neo4j GraphAcademy
- Machine Learning in Production - Coursera: DeepLearning.AI
- Supervised Machine Learning: Regression and Classification - Coursera: DeepLearning.AI
- Advanced Learning Algorithms - Coursera: DeepLearning.AI
- Analyze Data to Answer Questions - Coursera: DeepLearning.AI
- Ask Questions to Make Data-Driven Decisions - Coursera: Google Career Certificates
- Prepare Data for Exploration - Coursera: Google Career Certificates
- Process Data from Dirty to Clean - Coursera: Google Career Certificates
- Foundations: Data, Data, Everywhere - Coursera: Google Career Certificates

References

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